

Large Language Models and Knowledge Graphs Synergistically Enhancing Personalized Learning

Yuying Peng*

Department of Software and Big
Data Technology
Dalian Neusoft University of
Information
Dalian, China
pengyuying@neusoft.edu.cn

Siqu Zhao

Department of Software and Big
Data Technology
Dalian Neusoft University of
Information
Dalian, China
zhaosiqi@neusoft.edu.cn

Ximeng Luo

Department of Software and Big
Data Technology
Dalian Neusoft University of
Information
Dalian, China
luoximeng@neusoft.edu.cn

Yongzhi Ning

Department of Software and Big
Data Technology
Dalian Neusoft University of
Information
Dalian, China
873638313@qq.com

Abstract—The rapid development of artificial intelligence technology has promoted the intelligence of the education field. The collaborative work of large language models (LLMs) and knowledge graphs (KGs) can effectively develop personalized learning paths. Personalized learning in programming courses is crucial for meeting students' different needs and proficiency levels. This article proposes a method for personalized teaching of programming courses using LLMs and KGs. The LLMs is used to analyze students' learning status, provide real-time feedback, and generate customized learning materials, while the KGs provides a structured knowledge base to plan essential skills and learning progress, highly integrating the advantages of the LLMs and the logical connections of knowledge from various disciplines. The application results show that designing a knowledge system ontology through disciplinary knowledge analysis, retraining and fine-tuning the LLMs, forms the final personalized knowledge system, improves the accuracy of learning path recommendations and the relevance of generated exercises, and can be easily promoted to other disciplinary fields.

Keywords—large language models, knowledge graphs, personalized learning, artificial intelligence technology, programming courses

I. INTRODUCTION

With the rapid development of artificial intelligence technology, the education sector is undergoing an unprecedented transformation. As a learner-centered teaching concept, personalized learning [1] emphasizes providing customized learning experiences based on each student's unique needs, interests, learning styles, and ability levels. The implementation of this concept can not only significantly improve learning efficiency and effectiveness but also stimulate students' learning potential and promote their all-round development. However, the traditional education model faces numerous challenges in achieving personalized learning [2], such as the personalized adaptation of teaching resources, the dynamic adjustment of learning paths, and the accurate assessment of learning outcomes.

In recent years, the emergence of Large Language Models (LLMs) and Knowledge Graph (KGs) has provided new ideas and tools to address these challenges. LLMs [3-6] with its powerful natural language processing capabilities, can

understand and generate natural language text, providing personalized learning content and feedback for students. It can accurately grasp students' learning status and needs by analyzing their learning behavior, question answering, and text generation, and provide targeted learning advice and guidance. In addition, the big language model can also generate diverse learning materials, such as practice questions, case studies, and simulated dialogues, enriching students' learning experience. And the KGs [7-10] is a structured semantic knowledge base that can represent complex knowledge points and their interrelationships in the form of graphs. It provides a powerful knowledge management and organizational tool for the education field, which can help teachers and students clearly understand the logical structure of course content and optimize learning paths. Through KGs, students can quickly locate the knowledge points they need to learn and deepen their understanding by associating them with other knowledge points. Meanwhile, KGs can also support intelligent recommendation systems, recommending relevant learning resources and expanding content based on students' learning progress and interests.

LLMs with KGs can achieve complementary strengths and further enhance the effectiveness of personalized learning. The natural language processing capabilities of LLMs can strengthen the dynamic updating and content generation of knowledge graphs, enabling them to better adapt to students' learning changes [11]. Conversely, the structured knowledge of knowledge graphs can provide LLMs with richer background knowledge and logical frameworks, improving the accuracy of their reasoning and recommendations [12]. This synergistic effect not only offers students a more precise and efficient learning experience but also provides teachers with powerful instructional support tools to better understand and guide the learning process of students [13].

This paper proposes a mechanism for the collaborative application of LLMs and KGs in personalized learning, analyzing their specific implementation methods and effects in different educational contexts. By constructing a personalized learning system based on LLMs and KGs, this paper will verify its potential in enhancing learning outcomes, optimizing learning pathways, and enriching the learning experience. It will also provide theoretical support and practical guidance for the future development of educational technologies.

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II. RESEARCH FRAMEWORK AND EXPERIMENTAL DATA

A. Research Framework

This article focuses on personalized learning and constructs a personalized learning system based on the synergistic effect of Large Language Models (LLMs) and Knowledge Graph (KGs). Firstly, by analyzing disciplinary knowledge and designing the ontology of the knowledge system, a structured knowledge graph is constructed to provide a logical framework and knowledge foundation for learning path planning. Secondly, utilizing the powerful natural language processing capabilities of LLMs, real-time analysis of students' learning status can be conducted to generate personalized learning feedback and customized learning materials. On this basis, LLMs are retrained and fine tuned through technologies such as LoRA [14] to better adapt to personalized learning needs. The system combines the dynamic generation ability of LLMs with the structured knowledge of KGs to plan personalized learning paths for students and dynamically adjust them according to their learning progress. Finally, the effectiveness of the system in improving the accuracy of learning path recommendation, generating exercise relevance, and learning outcomes was verified through experiments, and its potential for promotion in other disciplinary fields was explored. The research framework of this article is shown in Figure 1.

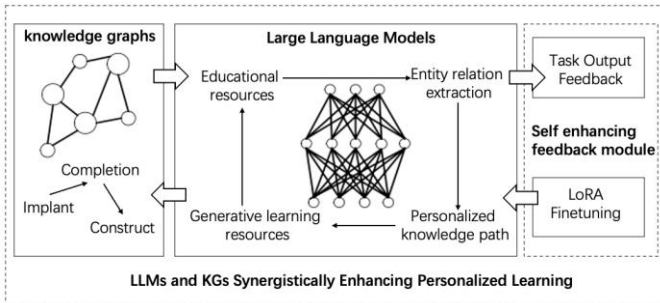


Fig. 1. The research framework

B. Experimental Data

The experimental data includes the learning data of 200 software engineering students from different grades and learning levels to ensure diversity and representativeness of the data. The dataset covers multidimensional information of students in programming courses, including weekly learning time, frequency of program code submission, homework completion status, classroom participation, and interaction records on online learning platforms. In addition, self-evaluation of students' learning preferences, programming skills, and feedback on the use of different teaching resources were also collected. These data are obtained through various methods such as questionnaire surveys, programming competition platforms, and teacher evaluations, aiming to comprehensively reflect students' learning behaviors and needs, and provide rich data support for the planning of personalized learning paths and the generation of learning materials.

III. ALGORITHM IMPLEMENTATION

This algorithm aims to assist teaching by accurately analyzing personalized data of students through the system. Firstly, the KGs is constructed through disciplinary knowledge

analysis to clarify the course knowledge points and their logical relationships, forming a structured knowledge system. Then, using the LLMs to analyze students' learning behavior data (such as learning time, homework completion status, etc.), generate a portrait of students' learning status. Based on this portrait, the system fine tunes LLMs using LoRA technology to generate personalized learning paths and materials for each student. At the same time, the system combines the knowledge structure in the knowledge graph to plan the learning path for students, ensuring the logical coherence of the learning content. During the learning process, the system continuously collects student feedback, dynamically adjusts the output of LLMs, and optimizes the relevance of learning paths and materials. Ultimately, by comparing students' learning outcomes and satisfaction, the effectiveness of the system is verified, and the algorithm is further optimized based on the evaluation results, thereby helping teachers better understand students' performance in the classroom and achieve more personalized teaching and evaluation. The algorithm flowchart is shown in Figure 2.

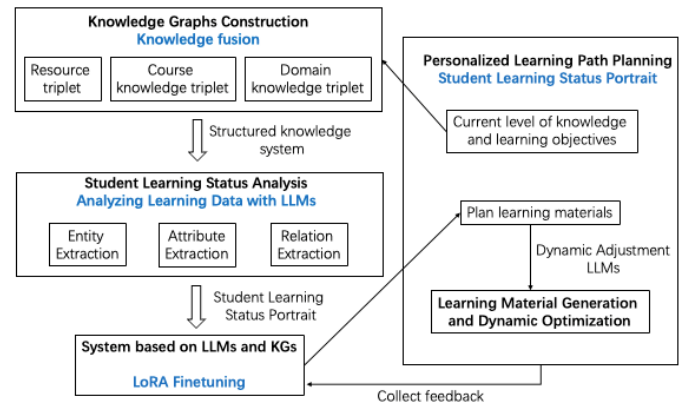


Fig. 2. The algorithm flowchart

A. Knowledge Graph Construction

In the development of personalized learning systems, the first step is to build a structured knowledge graph (KG) to comprehensively reflect the knowledge system and logical relationships of programming courses. By analyzing the course outline, textbook content, and teaching objectives in depth, identify key knowledge points (such as data structures, algorithm design, programming language features, etc.) and define their relationships (such as prerequisite relationships, inclusion relationships, etc.). Use tools to organize these knowledge points and relationships in the form of triplets, forming a hierarchical knowledge graph. In addition, supplementary knowledge is extracted from multiple sources such as online course platforms, academic literature, and programming communities to enrich the content of the knowledge graph and ensure that it covers all core knowledge points and extended content of the course. Knowledge graph not only provides students with a clear knowledge navigation framework, but also lays the foundation for personalized learning path planning and material generation in the future. As shown in Figure 3.

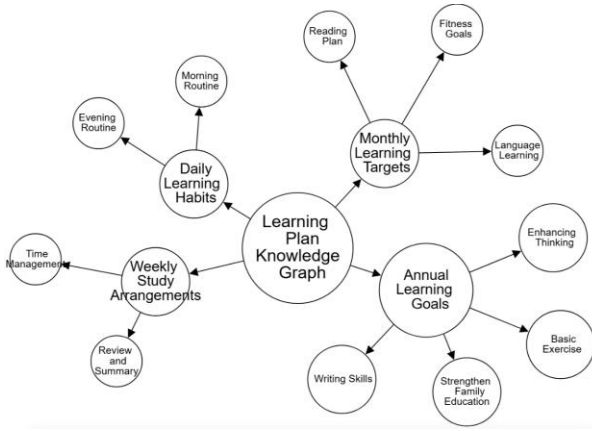


Fig. 3. representation of Knowledge Graph Section

B. Student Learning Status Analysis

In order to achieve personalized learning guidance, the system needs to accurately analyze the learning status of each student. By collecting multidimensional data from students during the learning process, including weekly study time, code submission frequency, homework completion quality, classroom participation, and interaction records on online learning platforms, a student learning behavior profile is constructed. Utilizing the powerful natural language processing capabilities of Large Language Models (LLMs), text analysis is conducted on students' programming assignments, online discussions, and problem feedback to extract their mastery of knowledge points and learning difficulties. Based on students' self-assessment and teachers' evaluation, the system generates a comprehensive learning status report, covering students' current knowledge level, learning style, and interest preferences. This analysis provides important input basis for subsequent personalized learning path planning and learning material generation.

C. Personalized Learning Path Planning

Based on the analysis of students' learning status and structured KGs, the system utilizes LLMs to plan personalized learning paths for each student. Firstly, based on students' current knowledge level and learning objectives, relevant knowledge points and their logical relationships are extracted from the KGs, to determine the core content and sequence that students need to learn. Combining the reasoning ability of LLMs, predict the difficulties that students may encounter during the learning process, and plan auxiliary learning materials and exercises in advance to help students better grasp knowledge. At the same time, the system dynamically adjusts the teaching methods and resource types in the learning path based on students' interests, preferences, and learning styles, ensuring that the learning path not only conforms to the course logic but also meets students' personalized needs. In this way, the system generates an efficient and targeted learning path for each student, helping them systematically improve their programming skills.

D. Learning Material Generation and Dynamic Optimization

At the same time as developing personalized learning paths, the system utilizes LLMs to generate learning materials that match the learning path. Based on the description of knowledge

points in the knowledge graph and students' learning status, LLMs generate diverse learning materials such as exercises, code examples, and case studies to enhance students' understanding and application abilities of knowledge points. At the same time, the system dynamically adjusts the LLM generation strategy by continuously collecting students' feedback on learning materials (such as completion time, accuracy, satisfaction, etc.), optimizing the difficulty, type, and content of learning materials. For example, if a student performs poorly in a practice question on a certain knowledge point, the system will generate more targeted coaching materials; If students have low satisfaction with a certain type of learning material, the system will adjust the form or content of the material. Through this dynamic optimization mechanism, the system ensures that learning materials are always aligned with students' learning progress and needs, thereby improving learning effectiveness and students' learning experience. The optimized structure of the model is shown in Figure 4.

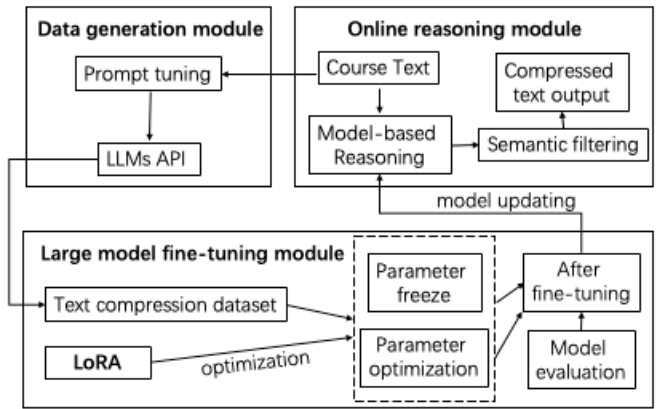


Fig. 4. Model optimization structure diagram

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Optimize Personalized Learning Path

The personalized learning system based on LLMs and KGs exhibits significant advantages in multiple key indicators. (1) The accuracy of learning path recommendations is outstanding, with the experimental group students' path matching rate increasing from 65% to 87%, far higher than the control group's 55%. This indicates that the system can dynamically adjust the recommendation path based on students' learning status and logical relationships in the knowledge graph, making it more in line with students' needs. (2) The learning materials generated by the system also performed well in terms of relevance and satisfaction. The experimental group had an 80% utilization rate of the learning materials, with a satisfaction score of 4.5 out of 5, while the control group had scores of 50% and 3.2, respectively. This indicates that the materials generated by the system are not only highly correlated with students' learning progress, but also effectively enhance their interest and participation in learning. (3) The average score of the experimental group students in the programming course increased from 72 points to 85 points, and the pass rate increased from 80% to 92%, while the average score of the control group only increased from 70 points to 75 points, and the pass rate increased from 78% to 82%. The results indicate that the system

significantly improves students' learning outcomes through precise learning path recommendations and high-quality learning material support, especially in programming courses that require highly personalized guidance. The experimental results are shown in Figure 5.

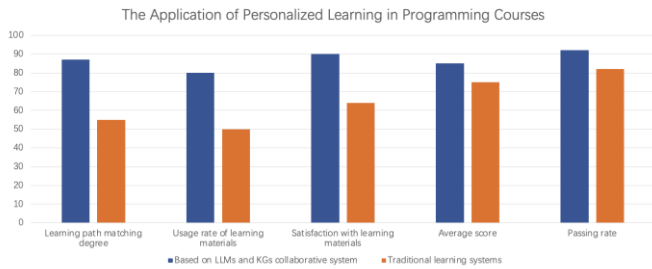


Fig. 5. Experimental data analysis

B. Optimize the Digital Evaluation System

By generating precise learning paths and high-quality learning materials, students' learning outcomes in programming courses have been significantly improved, providing a richer data foundation for the digital evaluation system. The system can not only track students' learning progress and knowledge mastery in real time, but also generate detailed evaluation reports through the interaction data between students and learning materials (such as exercise completion rate, code submission quality, etc.). These reports not only reflect students' learning outcomes, but also reveal strengths and weaknesses in the learning process, providing teachers with more comprehensive feedback and helping them adjust their teaching strategies. By utilizing the natural language processing capabilities of LLMs, it is possible to automatically grade and provide feedback on students' programming assignments and projects, further improving the efficiency and accuracy of evaluation. Through this comprehensive digital evaluation system, teachers can more accurately assess students' learning status, identify potential problems in a timely manner, and provide targeted guidance, thereby promoting the continuous improvement of teaching quality and student learning outcomes. As shown in Figure 6.

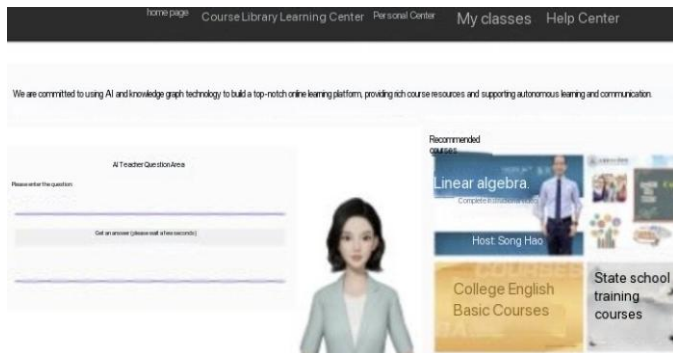


Fig. 6. Screenshot of system recommendation page

C. Application in the Field of Educational Technology

In programming courses, the system can dynamically generate personalized code exercises and project tasks based on

students' current programming skills and learning progress, while providing real-time feedback and code optimization suggestions to help students gradually improve their programming abilities. In online learning platforms, the system analyzes students' learning behavior and knowledge mastery, intelligently recommends course resources and learning paths that match their level, significantly improving the utilization of learning resources and students' learning efficiency. In intelligent tutoring systems, LLMs combined with KGs can provide students with accurate explanations of knowledge points and problem-solving, and even generate simulated dialogue scenarios to help students better understand complex concepts. For example, in Java course learning, the system can generate targeted problem-solving strategies and practice questions based on students' grammar error types. These applications not only enhance students' learning experience, but also provide powerful tools for educators to better understand students' needs, optimize teaching content and methods, and promote the development of educational technology towards a more intelligent and personalized direction.

V. CONCLUSION

This article proposes a personalized learning method based on the synergistic effect of Large Language Models (LLMs) and Knowledge Graph (KGs), aiming to enhance the teaching effectiveness of programming courses and students' learning experience through intelligent means. This method constructs a structured knowledge graph, clarifies course knowledge points and their logical relationships, and uses LLMs to analyze students' learning status, generating personalized learning paths and materials. By using LoRA technology to fine tune LLMs, the system can dynamically adjust the learning content to ensure a high match with students' learning progress and needs. The experimental results showed that the system significantly improved the accuracy of learning path recommendations, the relevance of learning materials, and students' academic performance. The average score of the experimental group students increased from 72 points to 85 points, and the pass rate increased from 80% to 92%.

This study demonstrates how to use artificial intelligence technology and knowledge management tools to support blended learning environments. By combining the two, the shortcomings of knowledge graphs can be compensated for, such as automatic completion and expansion of knowledge graphs, understanding context and implicit knowledge, and integrating multimodal data. The big language model provides more comprehensive and accurate causal inference through semantic understanding and reasoning abilities, solving the problem of personalized learning that is difficult to achieve in traditional education models. By accurately identifying students' learning needs and feedback, teachers can better grasp the classroom atmosphere, adjust teaching strategies in a timely manner, and thus improve students' learning experience and overall teaching quality. In the future, we will also explore the application effects of this system in more teaching environments, including distance education, special education, and lifelong learning, to promote the development of educational technology towards a more intelligent and personalized direction.

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